

Challenge 2 — Data Card

Project: Optimizing RAG Document Retrieval for Agronomic Advice

Team: Marula

Dataset Overview

Dataset component	Quantity / description
Agricultural documents	695
Training queries	308
Query-document relevance judgments (qrels)	4,194
Test queries	200
Document text length	Average 66.1 words; range 20–180 words
Training query length	Average 9.4 words; range 6–16 words
Relevance scale	0–3

1. Input vs. Output

Input: The system receives a farmer or agricultural information need as text. The document corpus is also text-based and contains agricultural document fields including document ID, title, text, source, crop, country, origin, source URL, and license.

Output: The retrieval system ranks the documents and returns the five most relevant document IDs for each query. The competition submission records the query ID and document ID in rank order. The task is retrieval and ranking rather than generation of an agricultural answer.

2. Classification Task

This project is not a conventional classification task. Instead, it is a document retrieval and ranking task. For the training data, query-document pairs have graded relevance labels from **0 to 3**, where higher values indicate greater relevance. The relevance distribution in the provided qrels is: **2,724** judgments at 0, **728** at 1, **186** at 2, and **556** at 3. These labels are used to evaluate how well the retrieval system ranks relevant documents, using nDCG@5.

3. Labelling Strategy

The relevance labels come from the competition-provided **qrels_train.csv** file. The team did not create a new human-labelling exercise for this project. The provided labels assign a relevance value from 0 to 3 to query-document pairs. The team checks the dataset schema, verifies that relevance values fall within the permitted 0–3 range, and uses the provided relevance judgments for local nDCG@5 evaluation. The team also compares retrieved documents against the available relevance judgments when analysing poorly performing queries.

4. Data Creation

The dataset used for the project is the dataset supplied through the Agricultural Extension RAG: Smart Retrieval for Farmers Kaggle competition. The provided files include **documents.csv**, **train_queries.csv**, **qrels_train.csv**, **test_queries.csv**, and the sample submission format. The document corpus contains 695 records. The inspected document records include source, country, crop, origin, source URL, and license information. The records shown in the project include documents marked as **synthetic** with **synthetic (CC0)** licensing. The project does not add manually collected personal data to the competition dataset.

5. Web Collection Ethics

The team is not conducting additional web scraping for this project. The project uses the competition-provided dataset and permitted pretrained models attached through Kaggle. Because the team is not collecting new web data from individuals, it is not conducting a separate consent or participant-recruitment process. For the supplied documents, the team relies on the source and license information included in the dataset and does not attempt to obtain private information or hidden competition answers. The competition guidance also prohibits using hidden solution files, manually encoding test answers, or copying another participant's private work.

6. Bias and Representation

The 695-document corpus may not cover every agricultural region, crop, country, language, farming practice, or type of agricultural question. The dataset therefore should not be assumed to represent the full range of agricultural knowledge or all farming communities. The document length is also limited in this corpus, with an average of 66.1 words and a range of 20–180 words. The 308 training queries, averaging 9.4 words, represent a limited set of information needs. The relevance labels are unevenly distributed, with 2,724 judgments labelled 0 and 556 labelled 3.

To address these limitations during development, the team will inspect poorly performing training queries, compare different lexical and semantic retrieval methods, and evaluate whether changes improve retrieval quality rather than assuming that a method is better. Broader evaluation with appropriate agricultural experts and intended users would be needed before real-world deployment.

7. Values and Community Alignment

The project is designed to improve access to relevant agricultural information for smallholder farmers, agricultural advisers, and agricultural extension workers. The key values guiding the dataset and system design are **usefulness, relevance, transparency, responsibility, and accountability**.

The dataset supports these values by enabling the system to retrieve supporting agricultural documents and rank the most relevant documents near the top. The quantitative evaluation makes the technical objective measurable: the initial TF-IDF baseline achieved an nDCG@5 score of **0.5131** on the training queries, while the dense retrieval experiment achieved **0.7856**. These experiments show the importance of improving retrieval quality rather than simply matching individual words.

The team will document the data sources, retrieval methods, evaluation process, experiments, and known limitations. The current dataset and evaluation do not establish that the system will work equally well for every agricultural community or situation. Before real-world use, further evaluation with appropriate agricultural experts and intended users would be necessary.

8. Dataset Summary (150–250 words)

Our dataset is a text-based agricultural information retrieval dataset for the project “Optimizing RAG Document Retrieval for Agronomic Advice.” It contains 695 agricultural documents, 308 training queries, 4,194 query-document relevance judgments, and 200 test queries provided through the Kaggle competition. The document text contains an average of 66.1 words per document, with document lengths ranging from 20 to 180 words. Training queries contain an average of 9.4 words, ranging from 6 to 16 words. The documents include fields such as document ID, title, text, source, crop, country, origin, source URL, and license. Relevance judgments use scores from 0 to 3, with 2,724 judgments at 0, 728 at 1, 186 at 2, and 556 at 3. The system receives an agricultural question as text and retrieves and ranks five documents from the corpus. This is a retrieval and ranking task rather than a conventional classification task. Retrieval quality is evaluated using nDCG@5, which considers both document relevance and ranking position.

9. Ethics and Representation (Reflection Response)

The project uses the dataset supplied through the Kaggle competition and does not involve collecting new personal information from individuals. The team is not conducting additional web scraping for this project. The provided document records contain source and license information, and the records inspected by the team include documents

identified as synthetic and CC0. Representation remains an important limitation because the 695-document corpus may not cover every agricultural region, crop, country, language, farming practice, or type of agricultural question. The 308 training queries also represent only a limited set of information needs. The team will examine poorly performing queries and compare retrieval approaches. Broader data and evaluation involving appropriate agricultural experts and intended users would be needed before real-world deployment.

10. Values and Community Impact (Reflection Response)

The dataset supports the project's aim of improving access to relevant agricultural information for smallholder farmers, agricultural advisers, and agricultural extension workers. The key values guiding the design are usefulness, relevance, transparency, responsibility, and accountability. The quantitative evaluation connects these values to the technical objective: the initial TF-IDF retrieval baseline achieved an nDCG@5 score of 0.5131 on the training queries, while the dense retrieval experiment achieved 0.7856. The system therefore needs to place more relevant agricultural documents within the top five positions. The team will document its data sources, retrieval methods, evaluation process, experiments, and limitations. Before real-world use, further evaluation with appropriate agricultural experts and intended users would be necessary.

Note: Quantitative values in this Data Card are based on the dataset inspection and experiments provided by the team. Word counts are based on whitespace-separated word counts in the notebook; no token counts have been inferred or invented.